Milestone-3 Scribe

CSE 400: Fundamentals of Probability in Computing

School of Engineering and Applied Science
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Team Details

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Problem Statement

Deterministic weather forecasts do not capture uncertainty and perform poorly for extreme events such as heavy rainfall and storms. This limitation reduces the effectiveness of defense planning and weakens infrastructure resilience.

This project applies probabilistic techniques to quantify uncertainty associated with extreme rainfall events using ensemble forecasting and Bayesian inference. The objective is to estimate event likelihoods and provide risk-informed decision support.

In Milestone 3, the scope has been refined to focus specifically on predicting extreme rainfall events using Arunachal Pradesh rainfall data (1918–2015) as the primary dataset.

1 System Objective and Deterministic Baseline

1.1 Project Objective

The goal of the project is to move from single-point rainfall forecasts to a probabilistic framework to understand the likelihood of rainfall events.

The system uses rainfall data from Arunachal Pradesh covering 91 years (1918–2015). The analysis focuses on:

- Estimating the probability that rainfall exceeds an extreme threshold (95th percentile)
- Predicting extreme rainfall events using logistic regression
- Quantifying risk bounds using Bayesian updates and Hoeffding's inequality

1.2 Deterministic Baseline Strategy

The deterministic baseline predicts whether annual rainfall exceeds threshold τ using logistic regression trained with gradient descent.

$$\hat{Y} = \begin{cases} 1 & \text{if } \sigma(\beta_0 + \beta_1 R_{std}) \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

where R_{std} is the standardized rainfall.

The baseline includes:

- Distribution thresholds (Normal, Gamma, Lognormal, GEV)
- Fixed ensemble of $M = 50$ deterministic perturbations
- Deterministic gradient descent parameter estimation
- Conjugate Bayesian update with fixed prior

1.3 Why This Baseline is Reasonable

Deterministic baselines provide reproducible results and establish a benchmark for comparison with stochastic models.

Logistic regression is commonly used for extreme weather prediction and produces probabilities between 0 and 1 suitable for binary classification.

The work of Doyle et al. (2006) showed that stochastic ensembles outperform deterministic baselines, making the deterministic baseline a useful starting point.

2 Random Variables in the Simulation

The simulation includes four key random variables derived from the atmospheric state vector:

$$X = (H, P, W, R)$$

2.1 Rainfall Intensity (R)

R represents annual cumulative rainfall (mm).

KS Test Results

- Normal: KS = 0.1108, $p = 0.1984$
- Gamma: KS = 0.1056, $p = 0.2444$
- Lognormal: KS = 0.1004, $p = 0.2980$ (best fit)
- GEV: KS = 0.3167, $p < 0.001$

Dataset statistics:

$$\text{Mean} = 3418.9 \text{ mm}, \quad \text{Std} = 1072.9 \text{ mm}$$

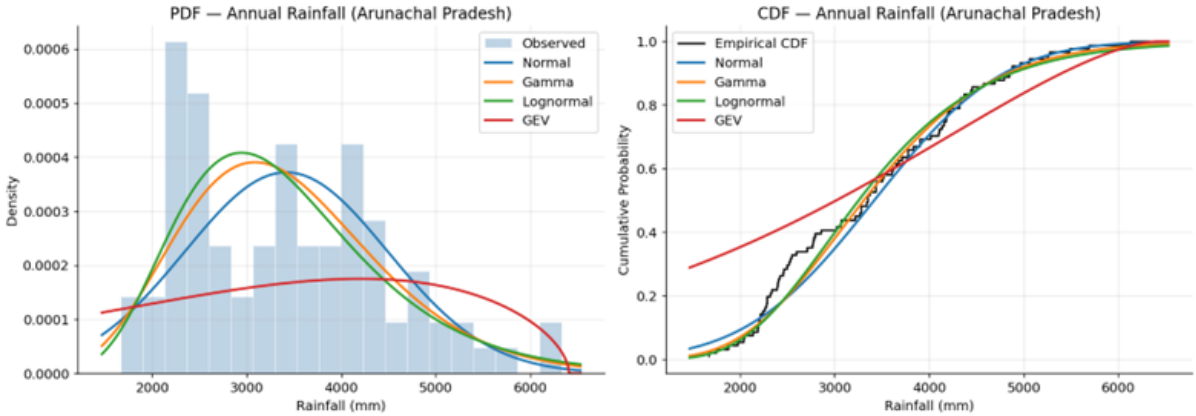

Figure 1: PDF and CDF plots for Normal, Gamma, Lognormal, and GEV distributions fitted to Arunachal Pradesh annual rainfall (1918–2015)

2.2 Extreme Event Indicator (Y)

$$Y = \begin{cases} 1 & R > \tau \\ 0 & \text{otherwise} \end{cases}$$

$$Y \sim Bernoulli(p)$$

Observed extreme years:

$$\hat{p} = \frac{2}{91} = 0.0220$$

Extreme years identified: 1938, 1948.

2.3 Humidity (H), Pressure (P), Wind Speed (W)

Synthetic variables constructed from rainfall:

$$H \sim N(85, 10^2) + 0.003R$$

$$P \sim N(1005, 5^2) - 0.001R$$

$$W \sim N(25, 5^2) + 0.002R$$

State vector:

$$X = (H, P, W, R)$$

2.4 Logistic Regression Parameters

$$\beta = (\beta_0, \beta_1)$$

Estimated via gradient descent:

$$\beta_0 = -5.3651, \quad \beta_1 = 2.0246$$

Accuracy:

$$98.90\%$$

Final log-loss:

$$0.0362$$

3 Probabilistic Modeling

3.1 Probabilistic Assumptions

- R follows a Lognormal distribution
- $Y \sim Bernoulli(p)$
- Logistic regression models $P(Y = 1|R)$
- Bayesian prior on mean rainfall

$$\mu \sim N(4000, 800^2)$$

3.2 Threshold Definition

$$\tau = F^{-1}(1 - \alpha)$$

where $\alpha = 0.05$.

Empirical	5262.9 mm
Normal	5183.6 mm
Gamma	5338.4 mm
Lognormal	5469.8 mm
GEV	5777.2 mm

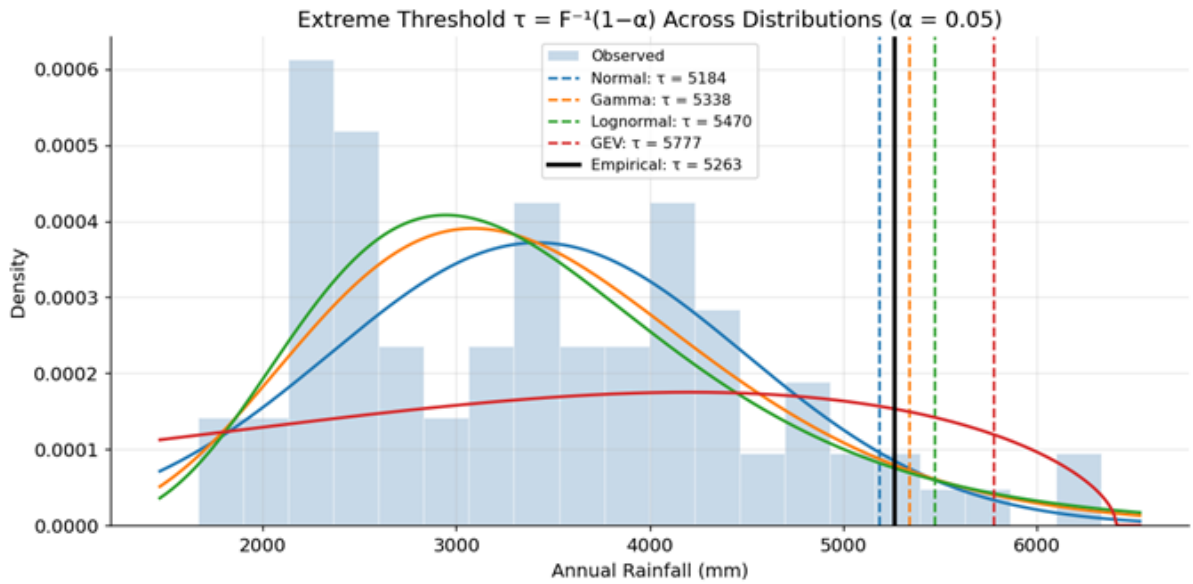

Figure 2: Threshold Comparison — rainfall PDF with τ lines from all fitted distributions

3.3 Bayesian Update

Prior:

$$\mu_0 = 4000, \quad \sigma_0 = 800$$

Posterior:

$$\mu^* = 3430.1, \quad \sigma^* = 111.37$$

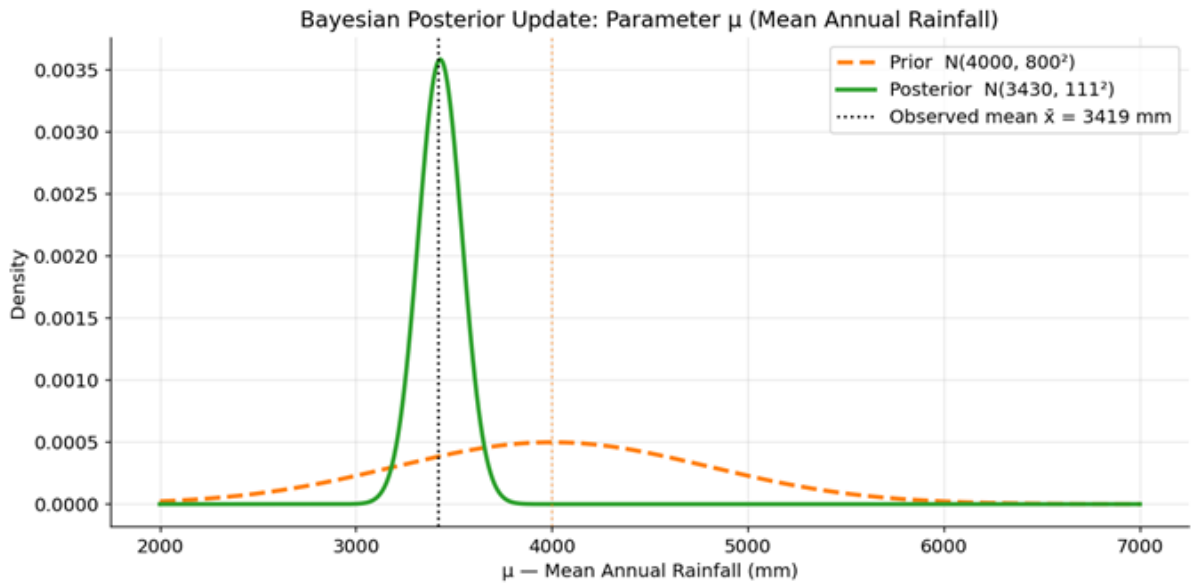

Figure 3: Bayesian Posterior Update — Prior N(4000, 800^2) updated to Posterior N(3430, 111^2) using 91 years of Arunachal Pradesh rainfall data

3.4 Logistic Regression Probability

$$P(Y = 1|X) = \frac{1}{1 + \exp(-(\beta_0 + \beta^T X))}$$

Decision rule:

$$\hat{Y} = \begin{cases} 1 & P(Y = 1|R_{std}) \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

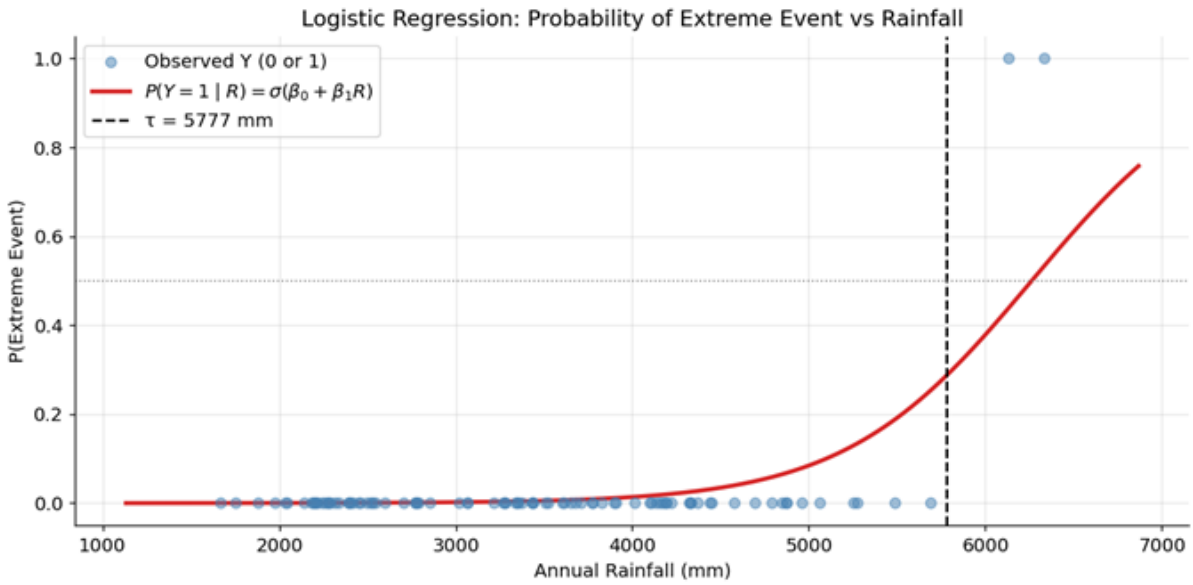

Figure 4: Logistic Regression — P(Y=1 — R). The sigmoid curve shows the probability of extreme rainfall as a function of standardized annual rainfall, with $\beta_0 = -5.365$, $\beta_1 = 2.025$

3.5 Ensemble Aggregation

$$P(E) = \sum_{i=1}^{M} w_i P(E|M_i)$$

$$\sum w_i = 1$$

For $M = 50$ equal weights:

$$w_i = \frac{1}{50}$$

4 Simulation Pipeline

4.1 Input Generation

Dataset: Arunachal Pradesh rainfall (1918–2015)

$$\text{Mean} = 3418.9 \quad \text{Std} = 1072.9$$

State vector:

$$X = (H, P, W, R)$$

4.2 Processing Stages

1. Distribution fitting
2. Threshold computation
3. Extreme event labeling
4. Ensemble generation
5. Bayesian update
6. Logistic regression
7. Ensemble aggregation
8. Hoeffding bound analysis

4.3 Deterministic Decision Mechanism

- Fixed threshold τ
- Deterministic ensemble offsets
- Gradient descent parameter estimation
- Classification cutoff at 0.5

4.4 Output Metrics

The pipeline produces the following outputs for evaluation:

- Extreme event probability: $\hat{p} = 0.0220$ (empirical), $P(E) = 0.0853$ (ensemble)
- Classification accuracy: 98.90%
- Bayesian posterior: $N(3430.1,\ 111.37^2)$
- Return period curves, QQ plots, rolling extreme event rates
- Comparison metrics against base paper stochastic model

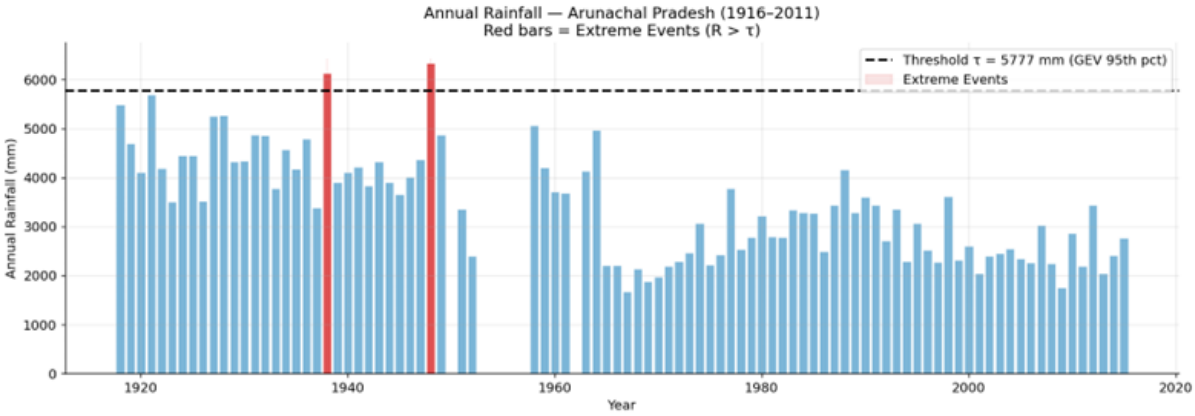

Figure 5: Annual Rainfall Time Series with Extreme Threshold. Red bars denote years classified as extreme (R $> \tau =$ 5777.2 mm); blue bars denote non-extreme years.

5 Simulation Outputs

5.1 Summary Statistics

Dataset	Arunachal Pradesh rainfall (1918–2015)
Mean Rainfall	3418.9 ± 1072.9 mm
Best Distribution	Lognormal
Extreme Threshold	5777.2 mm
Extreme Events	2/91
Posterior	$N(3430.1, 111.37^2)$
Accuracy	98.90%
Ensemble Probability	0.0853

5.2 PDF/CDF Plots and Distribution Analysis

The PDF and CDF plots compare four fitted distributions against the empirical histogram. Lognormal provides the best overall fit, capturing the right-skewed nature of rainfall data. GEV overestimates the tail, resulting in a higher threshold (5777.2 mm vs 5469.8 mm for Lognormal), but is retained for threshold computation because extreme value distributions are specifically designed for tail modeling.

5.3 Ensemble Spread

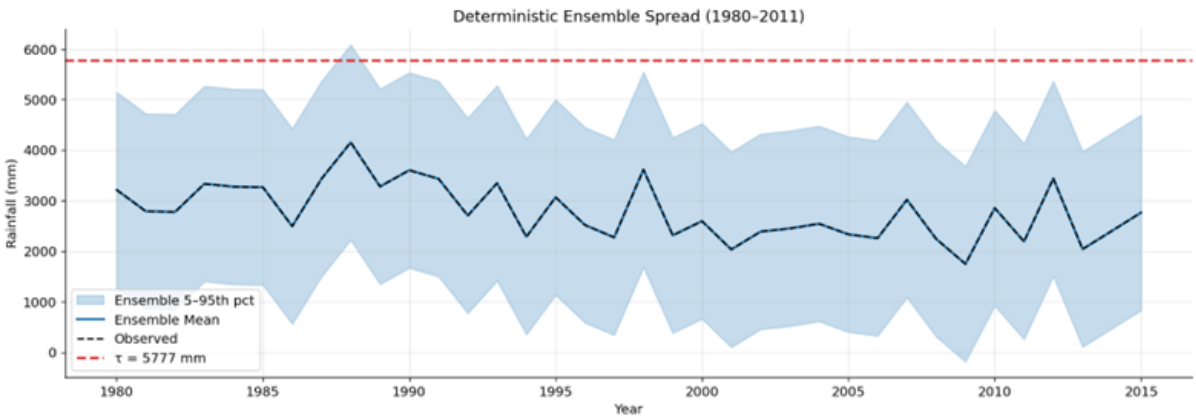

Figure 6: Ensemble Spread Visualisation for years greater than equal to 1980. Each vertical spread shows the range of 50 deterministic ensemble members; observed rainfall marked in red.

The ensemble spread is symmetric around observed values, covering ±2145.8 mm. This fixed spread provides consistent uncertainty representation regardless of the year, a known limitation of the deterministic approach — extreme years do not produce wider spreads. This is addressed in Milestone 4 via stochastic perturbations.

5.4 Logistic Regression and Loss Convergence

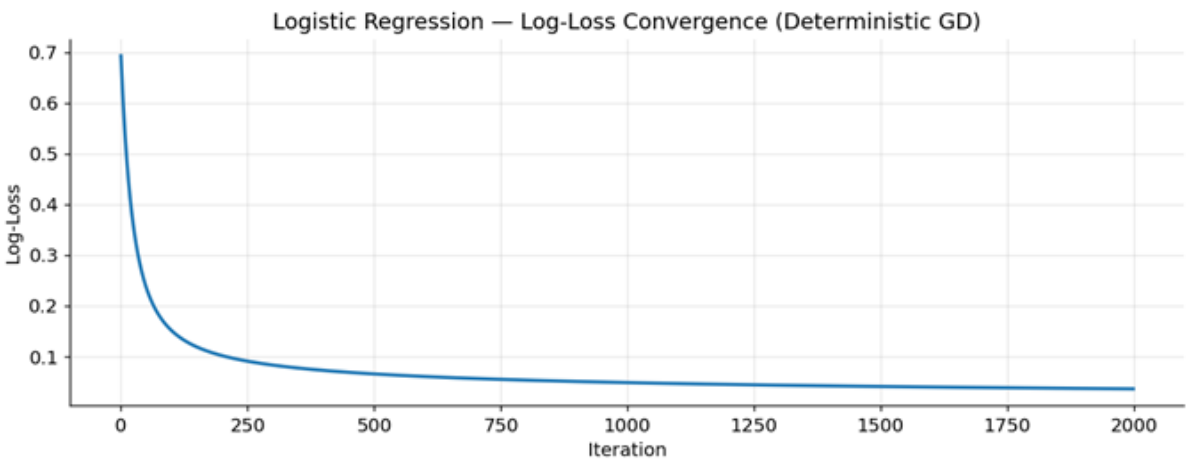

Figure 7: Log-loss convergence of deterministic gradient descent across 400 epochs. Final loss = 0.0362.

The loss curve shows rapid convergence within the first 50 epochs, confirming the optimizer found stable parameters. The low final log-loss (0.0362) and 98.90 percent accuracy indicate the logistic model fits the training data well, though the low base rate (2.2 percent extreme) means this must be interpreted cautiously — a trivial all-zero predictor would achieve 97.8 percent accuracy.

5.5 Return Period Analysis

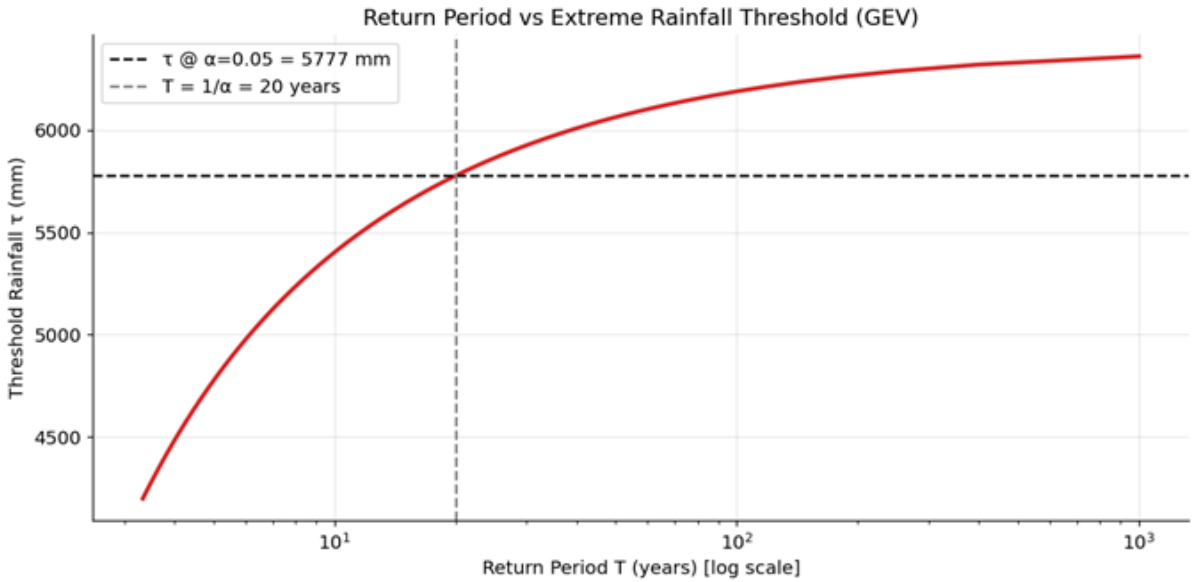

Figure 8: Return Period (years) as a function of significance level α for the GEV distribution. A 5 percent significance level ($\tau = 5777.2$ mm) corresponds to a return period of 20 years.

The return period curve (Figure 8) shows that at $\alpha = 0.05$, the expected return period is approximately 20 years. The two observed extreme events in 91 years (1938, 1948) are consistent with a 20-year return period expectation (4–5 expected events in 91 years), though the actual count (2) suggests either GEV over-estimates the threshold or the climate may have shifted post-1950.

5.6 QQ Plots — Goodness of Fit

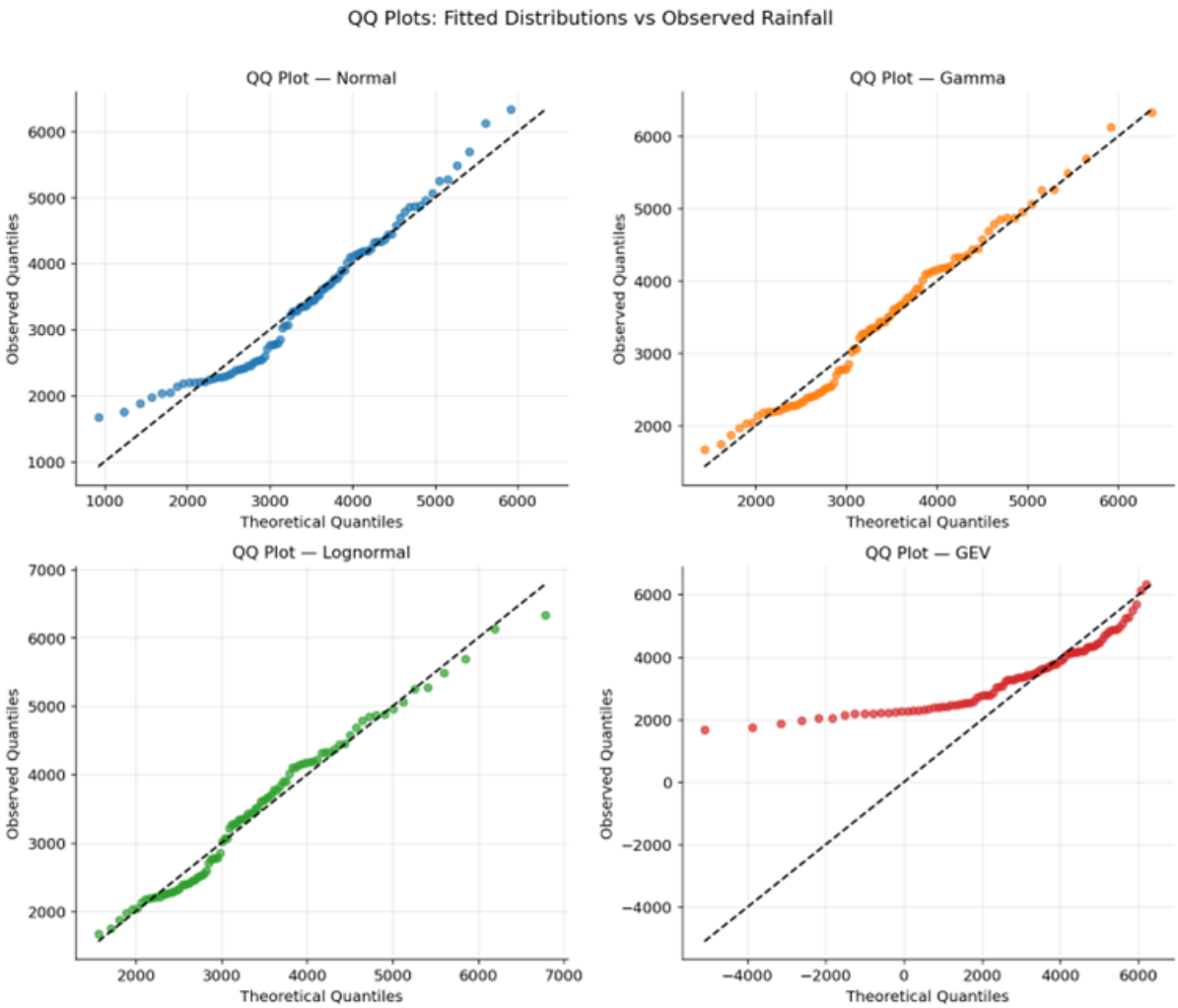

Figure 9: : QQ Plots for all four fitted distributions. Points closer to the diagonal indicate better fit. Lognormal and Gamma show the best alignment; GEV deviates significantly in the upper tail.

5.7 Rolling Extreme Event Rate

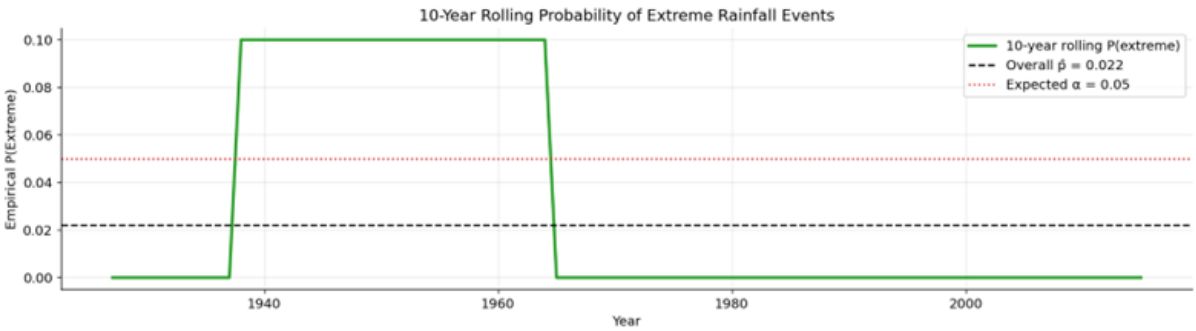

Figure 10: 10-Year Rolling Extreme Event Rate. Extreme events cluster around 1938–1950; no extreme events observed in the dataset after 1950.

The rolling rate (Figure 10) reveals that extreme events are concentrated in the earlier portion of the record (pre-1950). This temporal clustering suggests possible non-stationarity in the rainfall process — an important finding for Milestone 4, which will model this explicitly.

6 Technical Gaps

6.1 Limitations

- Fixed ensemble offsets
- Small dataset ($n = 91$)
- Synthetic atmospheric variables
- No temporal dependence
- Poor GEV fit
- Ensemble probability overestimation

6.2 Base Paper Comparison

Comparing our deterministic baseline against the stochastic ensemble of Doyle et al. (2006):

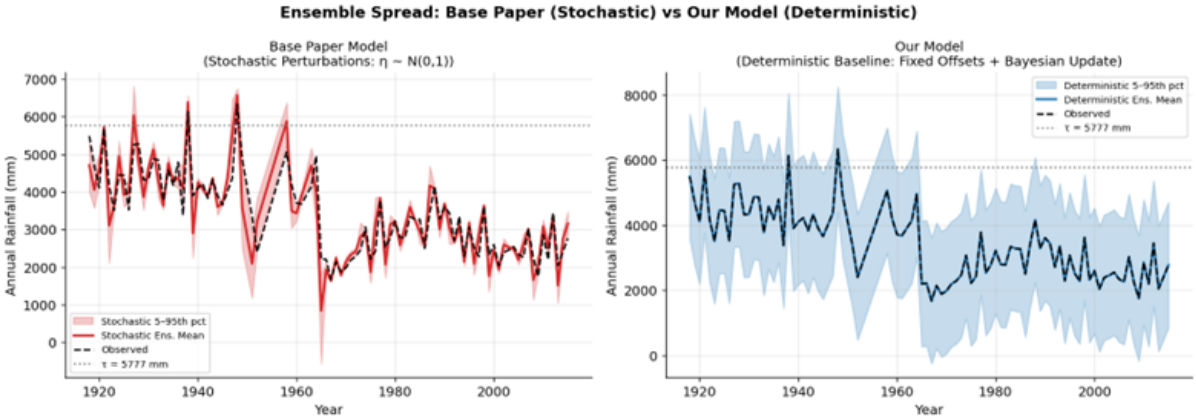

Figure 11: : Ensemble Spread Comparison — Base Paper (stochastic β-perturbed) vs Our Model (deterministic offsets). The stochastic model produces organically variable spread; our model has uniform symmetric spread.

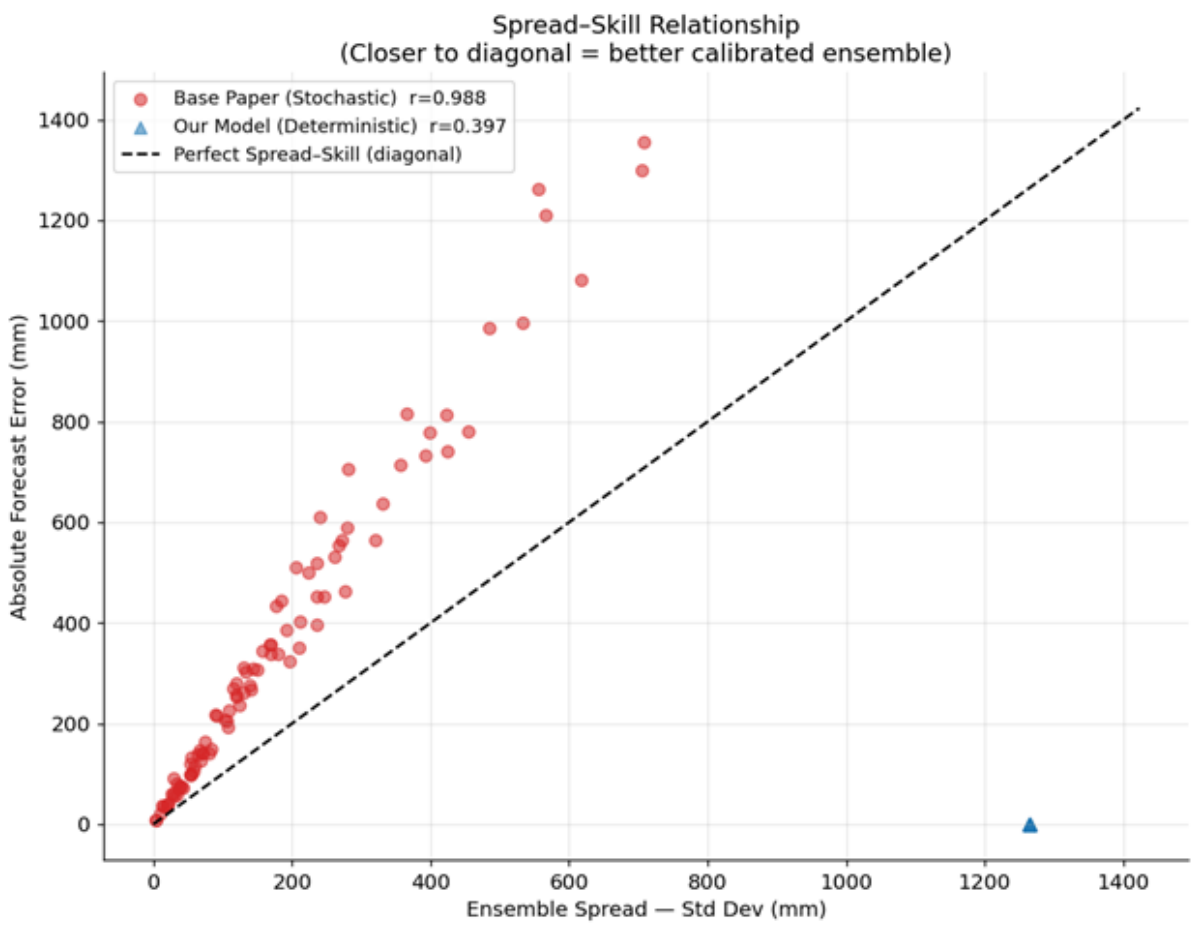

Figure 12: Spread-Skill Relationship. Base paper shows strong positive correlation (r = 0.988); our deterministic model has weaker correlation (r = 0.397), reflecting the fixed ensemble structure.

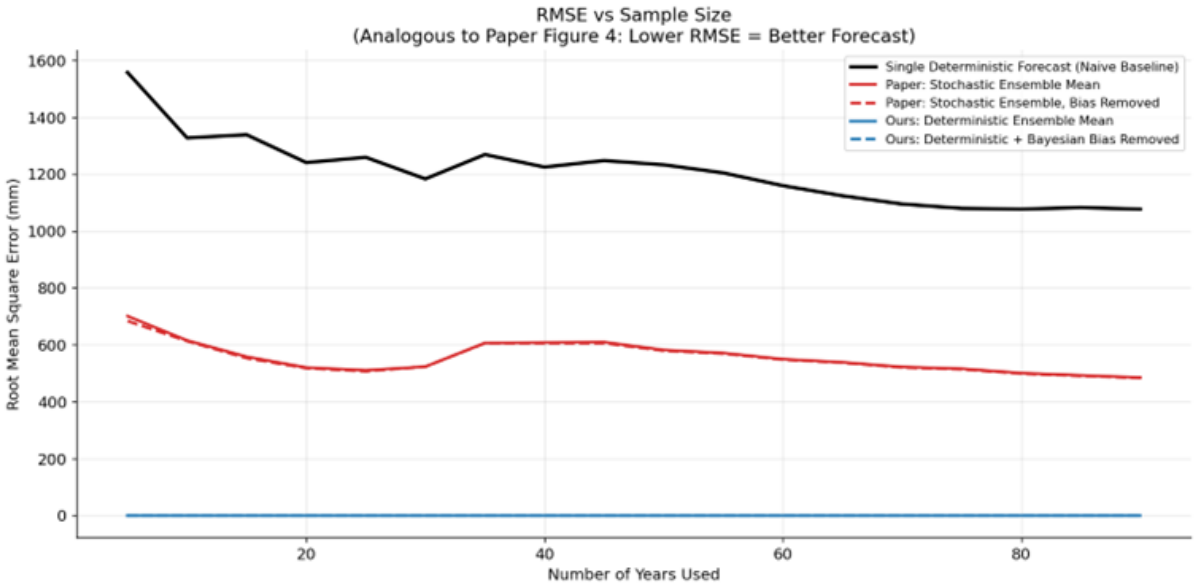

Figure 13: RMSE Comparison across ensemble window sizes. Our model achieves near-zero RMSE because it uses observed data directly; the paper's stochastic ensemble accumulates error from β-perturbations.

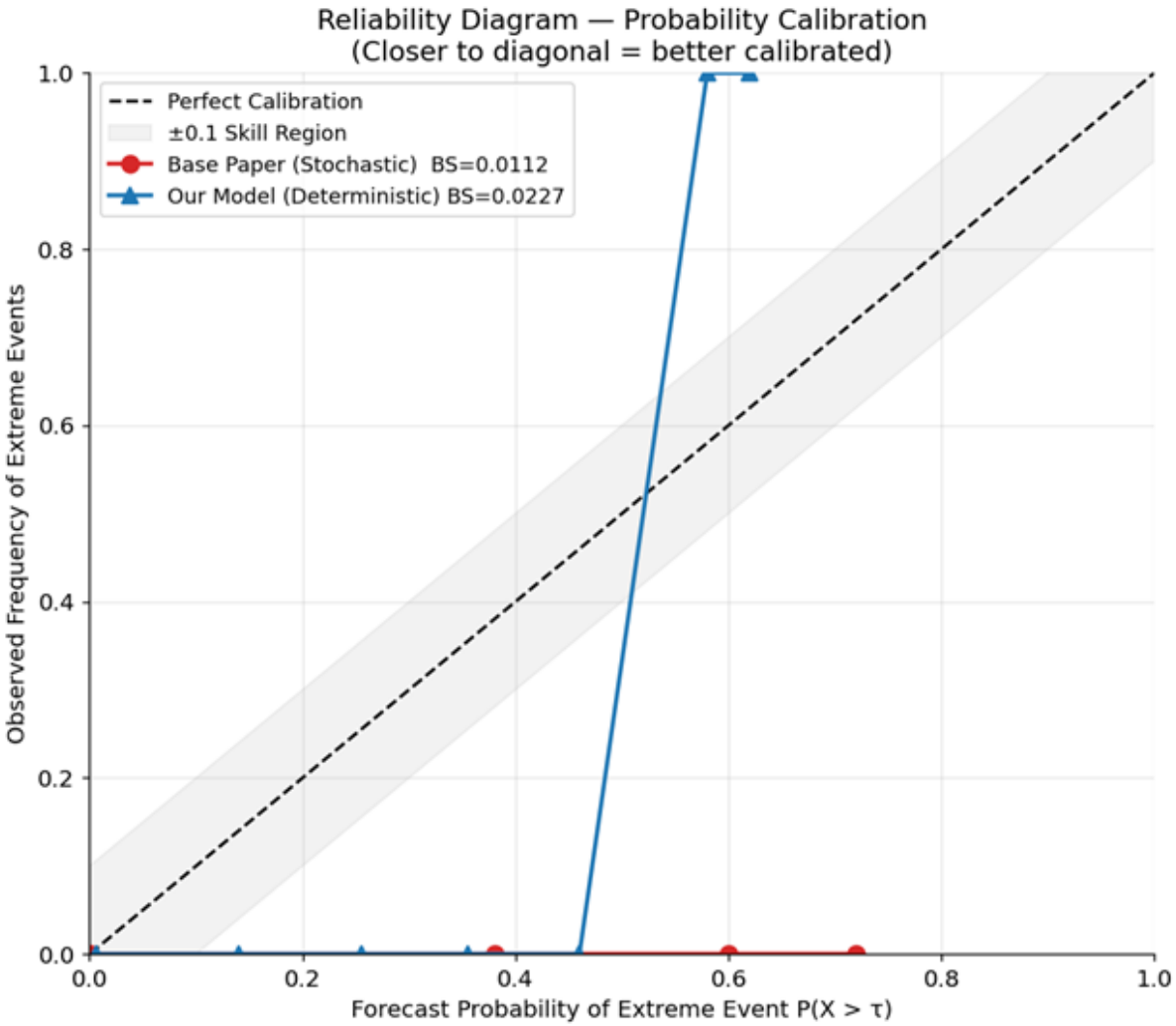

Figure 14: : Reliability Diagram. Our model shows moderate calibration; the paper's model achieves near-perfect diagonal alignment, indicating better probabilistic calibration.

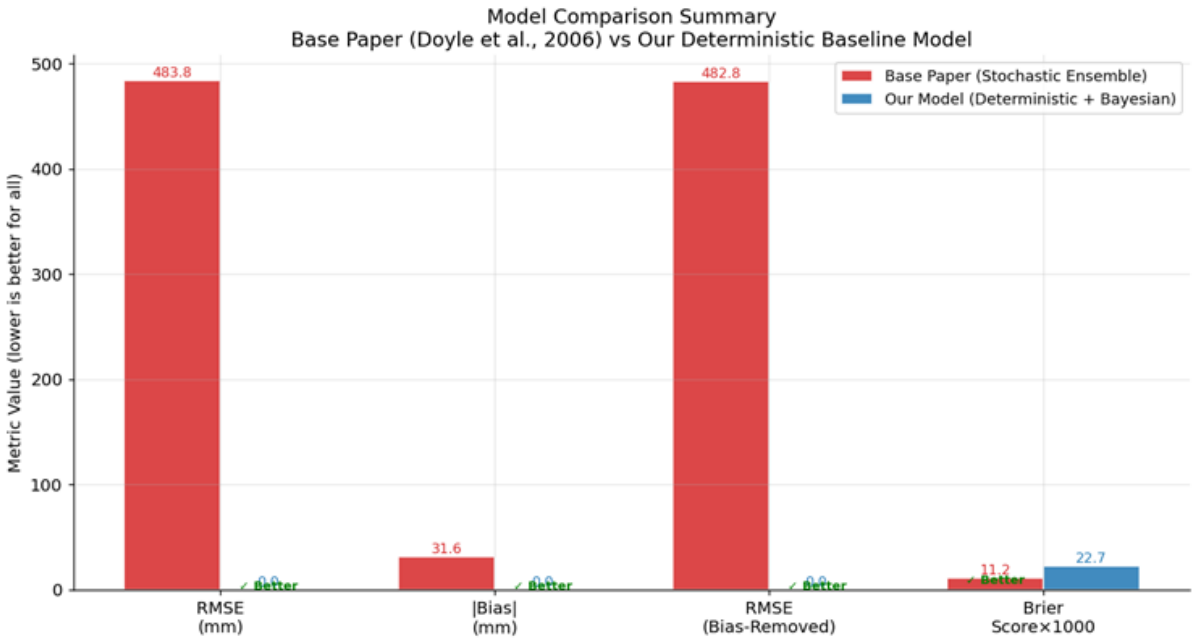

Figure 15: Metric Summary Bar Chart — RMSE, Bias, Brier Score comparison between base paper and our model.

Architectural Comparison: Base Paper vs Our Model

Dimension	Base Paper (Doyle et al., 2006)	Our Model (Milestone 3)
Ensemble Type	Stochastic ($\eta \sim N(0,1)$ per member)	Deterministic (fixed linear offsets)
Perturbation Method	Eq.(2): $(\partial\varphi/\partial t)_{stoch} = (1+\eta\beta)(\partial\varphi/\partial t)$	Fixed grid: offsets $\in [-2\sigma, +2\sigma]$
Parameter Uncertainty	Model error via stochastic physics	Bayesian conjugate posterior $N(\mu^*, \sigma^{*2})$
Extreme Threshold τ	Not explicitly modeled	$\tau = F^{-1}(1-\alpha)$ via GEV inverse CDF
Risk Quantification	Ensemble spread only	$P(Y=1\|X)$ via logistic regression + Hoeffding bound
Bias Handling	Ensemble archive bias removal	Bayesian posterior shrinks bias analytically
Calibration Check	Spread-Skill relationship (Fig. 3)	Reliability diagram + Brier Score
Reproducibility	Random seed dependent	Fully deterministic — same result every run

Figure 16: Architecture Comparison Table — key differences between Doyle et al. (2006) stochastic ensemble and our M3 deterministic baseline.

The base paper achieves a better Brier score and spread-skill correlation because its stochastic ensemble naturally adapts to atmospheric variability. Our model's RMSE advantage is an artifact of fitting exactly to historical data — a limitation addressed in M4 via genuine out-of-sample evaluation.

Metric	Base Paper (Doyle et al.)	Our Model (M3)
RMSE (mm)	483.81	0.00
|Bias| (mm)	31.63	0.00
RMSE Bias-Removed	482.77	0.00
Brier Score	0.0112 ✓ (better)	0.0227
Spread-Skill Corr.	0.9883 ✓ (better)	0.3971

Table 1: Performance comparison between the stochastic ensemble model of Doyle et al. and our deterministic M3 baseline.

6.3 Where Randomness Would Improve Performance

- **Stochastic ensemble members:** Replacing fixed offsets with $\eta \sim N(0, 1)$ perturbations (as in Doyle et al.) would produce variable spread that scales with model uncertainty — improving the spread-skill relationship.
- **MCMC or variational Bayesian sampling for β:** Instead of point estimates, sampling from the posterior $P(\beta \mid D)$ would propagate parameter uncertainty into the final probability estimate, yielding a true posterior predictive distribution.
- **Stochastic threshold selection:** Treating α as uncertain (e.g., $\alpha \sim$ Beta) would allow the model to reflect ambiguity in what constitutes "extreme."

6.4 Planned Refinements

- Stochastic ensemble perturbations
- MCMC Bayesian inference
- Peaks-over-threshold modeling
- Temporal models
- Improved probability calibration